

Champion freedivers can hold their breath for up to 11 minutes.

BREATHING UNDERWATER

OXYGEN TANKS

Oxygen is vital for humans to stay alive, so the ability to carry it around is crucial in places where humans can't breathe naturally, such as outer space and underwater. Oxygen gas can be stored in tanks at high pressures and supplied through tubes to the mouth. A device called a regulator helps deliver the oxygen at the right pressure to the user, making it safe to breathe. Oxygen cylinders are also used in hospitals for patients with breathing difficulties.



BREATHING IN SPACE

An astronaut carries an oxygen tank in a backpack, which connects to the helmet. The air on Earth is only 21 percent oxygen, with most of the rest being nitrogen gas. An astronaut's spacesuit is filled with almost 100 percent oxygen.

THE LUNGS

Humans get oxygen by breathing in (inhaling) and channelling air down to their lungs. There, oxygen is transferred into the blood, which then circulates it to the rest of the body. A waste gas, carbon dioxide, passes back into the lungs from the blood and is removed when we breathe out (exhale).

↓ AIR FLOWS IN ↑ AIR FLOWS OUT

Air moves through the windpipe.

The windpipe splits into smaller branches.

At the ends of the branches, gases pass between the lungs and the blood.

Oxygen is drawn in through the nose and the mouth.

Carbon dioxide is forced out.

